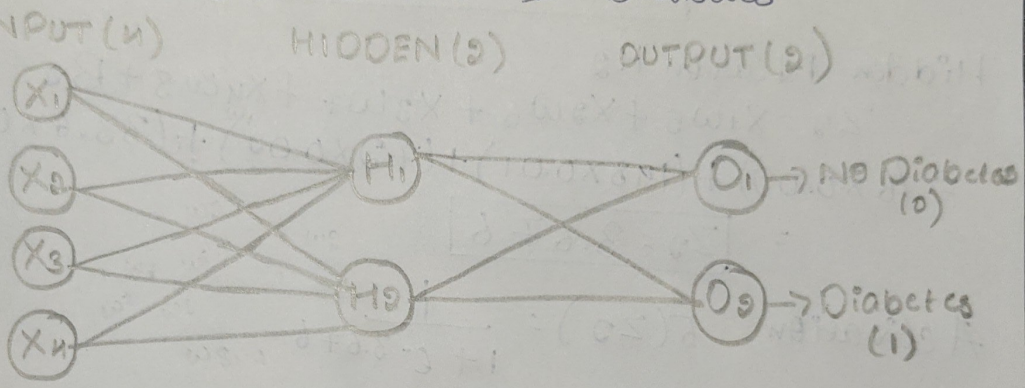
Keerthi.N

Assignment

The Pima Indians Diabetes Dataset is a classification dataset taken from Kaggle. It is used to predict whether a person has diabetes or not based on medical features such as preganancies, glucose, blood pressure and BMI. The dataset has 2 output classes : 0 - No Diabetes and 1 - Diabetes



Input - 4
- Pregnancies
- Glucose
- Blood Pressure
- BMI

Output → 2
- No Diabetes
- Diabetes

First Row $X = [6, 148, 72, 33.6]$

Actual output : Diabetes = 1

Hidden layer
Weights & Bias

For H_1 : $w_1 = 0.01$, $w_2 = 0.02$, $w_3 = 0.01$, $w_4 = 0.03$
$B_1 = 0.5$

For H_2 : $w_5 = 0.02$, $w_6 = 0.01$, $w_7 = 0.02$, $w_8 = 0.01$
$B_2 = 0.3$

Hidden layer H1

$$Z_1 = X_1 w_1 + X_2 w_2 + X_3 w_3 + X_4 w_4 + B_1$$

$= (6 \times 0.01) + (148 \times 0.02) + (72 \times 0.01) + (33.6 \times 0.03) + 0.5$

$= 0.06 + 2.96 + 0.72 + 1.008 + 0.5$

$Z_1 = 5.248$

Activation $\quad \sigma(z_1) = \frac{1}{1 + e^{-5.248}}$

$\boxed{H_1 \approx 0.9948 \simeq 1}$

Hidden Neuron H_2

$Z_2 = x_1 w_5 + x_2 w_6 + x_3 w_7 + x_4 w_8 + B_2$

$= (6 \times 0.02) + (148 \times 0.01) + (72 \times 0.02) + (33.6 \times 0.01) + 0.3$

$= \boxed{Z_2 = 3.676}$

Activation $\quad \sigma(z_2) = \frac{1}{1 + e^{-3.676}}$

$\boxed{H_2 \approx 0.9753}$

Output layer

$O_1 \rightarrow$ No Diabetes

$O_2 \rightarrow$ Diabetes

let: $\quad w_{H_1, O_1} = 0.4$

$w_{H_2, O_2} = 0.2$

$B_{O1} = 0.1$

$w_{H_1, O_2} = 0.3$

$w_{H_2, O_2} = 0.8$

$B_{O_2} = 0.1$

For No Diabetes

$Z_{\text{No Diabetes}} = H_1(0.4) + H_2(0.2) + 0.1$

$= (0.9948) \times 0.4 + (0.9753) 0.2 + 0.1$

$\boxed{Z = 0.6930}$

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For Diabetes

$$Z_{Diabetes} = H_1(0.3) + H_2(0.8) + 0.1$$
$$= (0.9948 \times 0.3) + (0.9753 \times 0.8) + 0.1$$
$$= 0.2984 + 0.7802 + 0.1$$

$$\boxed{Z_{Diabetes} = 1.1786}$$

Output Values

$$[0, 6930, 1.1786]$$

Since

$$1.1786 > 0.6930$$

The Highest output is Diabetes

Predicted output = Diabetes (1).